Titel: The Padma Corridor as a Spatial Transformer: Measuring Regional Accessibility

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Course: GEO 519

1. Introduction

Southern Bangladesh has long remained physically and economically detached from the central capital region due to the geographical barrier posed by the Padma River. For many years, people depended on slow ferry services to cross the river. The environmental impact assessment prepared for the Asian Development Bank explains that the region remained disconnected from the national road network because "the link between the southwest zone and the rest of the country across the Padma River is only by ferry" (Bangladesh Bridge Authority, 2010, p. 1-1). The ferry capacity was also limited; average waiting times sometimes exceeded two hours for buses and light vehicles and more than ten hours for trucks (Bangladesh Bridge Authority, 2010). These issues caused severe delays in passenger and freight movement. The drawbacks of these ferry services significantly restricted regional accessibility, economic development, and efficient connectivity with Dhaka and the rest of the country.

However, in 2014, the construction of the long-awaited Padma Bridge began, and it opened for public use in 2022. This corridor creates a direct and reliable road network connection between the southwestern region and the rest of the country. This significantly improves the transportation scenario of the landscape by reducing travel time and making travel more dependable (Geurs & van Wee, 2004). The bridge is having a positive impact on both transportation and economic growth. Fast and reliable access to the capital and ports is making the southwestern districts more

attractive for investment and industrial development. *The Business Standard*, a prominent business newspaper, estimated that the bridge could increase national GDP by 1.23% and boost the regional economy by about 2% (The Business Standard, 2022). This development has already started; two new Export Processing Zones (EPZs) have been established, and Ready-Made Garment (RMG) products now have a new route through Mongla Port (The Business Standard, 2022).

Besides that, Bangladesh is one of the most densely populated countries, and the urban population is increasing rapidly. The urban population increased from 14% in 1960 to 40.47% in 2023, while the rural population fell from 94.87% to 59.53% (Hamza, 2025). People who migrate for better opportunities often end up in capital cities; as a result. As a result, Dhaka has become one of the most densely populated cities in the world. The Padma Corridor has the potential to encourage industries to move away from the crowded capital city. Specifically, land-intensive industries, such as manufacturing and RMG factories, may relocate from Dhaka to new growth centers in the southwest region, which will result in reduced population concentration and ease the pressure on current infrastructure that is heavily focused on Dhaka.

The Padma Bridge is acting as a primary influencer for spatial change in southwestern Bangladesh by drastically increasing regional accessibility, fostering economic growth with a potential of balanced urban development. Moreover, accessibility is not just a simple measure of how easily or quickly one can travel from one place to another (Sundquist et al., 2020). As defined by Walter G. Hansen (1959), accessibility represents the "potential of opportunities for interaction" (Hansen, 1959, p. 73). Therefore, to measure the true impact of the Padma Bridge, an approach is required that accounts for both the friction of distance and the economic weight of the connected regions.

This study aims to answer: How has the Padma Bridge changed regional accessibility in southern Bangladesh between 2015 and 2025, and which upazilas have experienced the greatest changes?

2. Approach

2.2 Literature Review

To find out how Padma Bridge has altered the spatial distribution of regional accessibility, it is first necessary to define accessibility within the context of transportation geography.

Jean-Paul Rodrigue defines accessibility as "the capacity of a location to be reached" (Rodrigue, 2020). This means that spatial locations are inherently unequal. Different locations have different capacities. A location's capacity to facilitate the movement of people, freight, and information, making accessibility a key indicator of both spatial inequality and economic opportunity (Rodrigue, 2020).

Walter G. Hansen introduced a more comprehensive interpretation by defining accessibility as the *"potential of opportunities for interaction"* (Hansen,1959, p.73). Moving away from traditional connectivity or distance-based definition, Hansen emphasized that accessibility is not just about how easily a place can be reached, but also about what can be reached or the attractiveness of the locations. For example, from point A, if two locations have the same distance and travel time. The location which is economically vibrant and has more population will have the higher accessibility because it offers greater opportunities for employment, trade, and social interaction.(Hansen, 1959). In the context of transportation geography, every movement is

inherently purposive- people and goods travel to access opportunities that are unevenly distributed across space.

To effectively evaluate the spatial impact of the Padma Bridge, one must distinguish between geographic accessibility and potential accessibility. Geographic accessibility is a purely spatial metric; a straightforward summation of distances or travel times from point A to point B (Fig:1). The lower the calculated value, the more accessible a location is deemed to be. While this approach is useful for evaluating the raw physical connectivity of a network, it operates on the flawed assumption that all locations are of equal. On the other hand, potential accessibility is a socio-economically weighted metrics, that also takes other attributes of the location into consideration (Rodrigue, 2024).

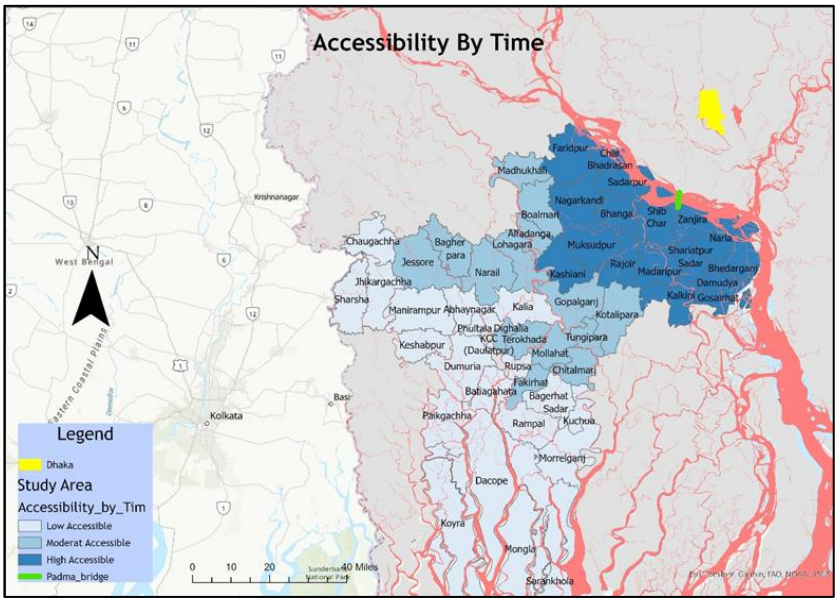


Figure-1: Accessibility by time

In reality, spatial interaction is driven by demand. As Rodrigue's concept of the "capacity to be reached" implies, all locations are not equal; some destinations are inherently more important or attractive than others due to the activities they have. Therefore, this study utilizes potential accessibility, which serves for a more complex and realistic measure. Potential accessibility goes beyond physical network by simultaneously including the concept of distance weighted by the specific attributes of a location, such as population density or economic growth.

Number of populations is often used in accessibility and spatial interaction studies as a proxy for the potential of interaction because larger populations generally generate greater economic, social, and transportation activities (Jacobs-Crisioni & Koomen, 2017). Therefore, population can be considered an effective indicator of destination attractiveness and interaction potential in accessibility analysis. As due to lack of number of jobs data, number of populations will act as proxy of opportunity of interaction. Night-time light (NTL) intensity has also become widely used in economic geography and regional studies as a proxy for economic growth and economic activity, especially in areas where reliable economic data are limited or unavailable. Satellite-observed night lights can effectively capture economic growth because increased economic activity generally leads to higher consumption of electricity and artificial lighting. Changes in light intensity strongly correlate with changes in GDP growth. (Henderson et al., 2009; Mellander et al., 2015). As we don’t have the micro level economic growth data this will act as economic proxy for our study.

2.2 Methodology

This study adopts a gravity- – based potential accessibility model to evaluate how the Padma Bridge changed regional accessibility in southern Bangladesh between 2015 and 2025. The model is based on Hansen's concept of accessibility as the "potential of opportunities for interaction," where accessibility not only depends on transportation network but also on the attractiveness of destinations. In this framework, accessibility increases with greater concentrations of population and economic activity and decreases with increasing travel friction.

To represent attractiveness, this study uses population and night-time light (NTL) intensity as proxy variables. Population is used as an indicator of potential social and economic interaction, while NTL intensity represents economic activity and urban development.

Internal road density was also considered to account for the quality of regional road connectivity. The regions with stronger internal transportation networks generally provide greater access to services, labor markets, and economic opportunities.

Travel friction was measured using travel time rather than physical distance. Travel time provides a more realistic representation of travel friction because it reflects actual transportation conditions, including road quality, congestion, and river-crossing constraints. A negative exponential decay function was applied to account the declining likelihood of trips as travel time increases. This approach follows accessibility literature suggesting that human travel behavior decreases nonlinearly with increasing travel cost and time.

Although accessibility is a fundamental concept in transportation and land-use planning, yet its practical application remains limited due to challenges in measurement and interpretation

and outcomes can vary significantly depending on how it is measured (Handy & Niemeier, 1997). So, it is importance to select appropriate indicators in analysis.

The accessibility model used in this study is expressed as:

$$\boldsymbol{A_i = RD_i \times (P_i^{\alpha} \times NTL_i^{\gamma}) \times e^{-\beta T_i}}$$

where:

- $\boldsymbol{A_i}$= potential accessibility of region i
- $\boldsymbol{ND_i}$= network density of region i
- $\boldsymbol{P_i}$= scaled population of region i
- $\boldsymbol{NTL_i}$= night-time light intensity of region i
- $\boldsymbol{T_i}$= travel time
- $\boldsymbol{\beta}$= travel friction parameter
- $\boldsymbol{\alpha}$**and** $\boldsymbol{\gamma}$= elasticity parameters for population and night-light intensity

Although the exponential decay constant should ideally be calibrated iteratively for the network rather than assumed directly, but due to limited resources meant the friction parameter (β) was set to 0.15, which falls within the commonly used range for regional-scale gravity-based accessibility models employing negative exponential travel-time decay functions (Geurs & van Wee, 2004b; R. W. Vickerman, 1974).

Equal weights were assigned to population and night-time light intensity (α=0.5, γ=0.5) because both variables represent complementary dimensions of regional attractiveness and there is no standard accepted theoretical basis for assigning a higher weight to either demographic or economic factors. Using equal weighting provides a balanced and transparent approach and also reduce subjective bias.

Accessibility values were calculated for both 2015 and 2025 to examine the spatial changes associated with the Padma Bridge corridor. The output was then compared to identify which upazilas experienced the greatest improvements in regional accessibility following the construction of the Padma Bridge.

Due to different kind of data source, workflow of this study relay relies on Google Earth Engine (GEE) for Remote Sensing Image analysis, ArcGIS for Spatial and Network Analysis, and RStudio for data management and accessibility calculation.

2.3 Data Sources

To accurately populate the parameters of the adapted Hansen gravity model, this study has to have demographic census data, geospatial data and data from remote sensing imagery.

Population data is from The Bangladesh Bureau of Statistics (BBS), which included the most recent demographic reports from 2022 as well as the original 2011 Population and Housing Census data.

Standardized shapefiles of administrative boundaries is downloaded from the Humanitarian Data Exchange (HDX).

Bangladesh Roads and Highways Department (RHD) GIS database is the source of network data to calculate the distance.

From these this raw data, other key variables were calculated for the final accessibility equation. Network density, representing the density of road, was calculated for each zone using ArcGIS. Network density is calculated by taking the total sum of road lengths within a administrative boundary and dividing it by the total area.

As there is no official data is available for speed limit and as traffic is not so fast is Bangladesh, a uniform travel speed of 60 km/hr. was assumed for all road segments. Furthermore, to accurately model the severe geographical barrier present in the 2015 pre-bridge scenario, a 4.5 hr. time penalty is added. There was an average 2 to 2.5-hour baseline waiting time for passenger vehicles and 2 hours for ferry (Development Bank, 2010).

A primary challenge in measuring potential accessibility across regional Bangladesh is the absence of high-resolution, uniformly surveyed economic data, such as Gross Domestic Product (GDP) at the local or upazila level. To accurately represent the comparative economic growth of a regions, this study uses remote sensing night-time light intensity as an proxy for economic growth. VIIRS Day/Night Band monthly composite collection satellite image was filtered according to study area for both 2015 and 2025 using Google Earth Engine. Then the mean light intensity was calculated for each administrative area or upazila.

3. Research Findings

The analysis of potential accessibility across the 58 upazilas for 2015 and 2025 shows a major spatial transformation across the southern regions of Bangladesh. The results indicate that reduced travel time played an important role in this change, but the transformation was not caused by travel time reduction alone. Instead, the large increase in accessibility resulted from the combined effect of other factors. When travel time (T) reduced, the effect of distance decay also became smaller, which increased the influence of local network connectivity (NC) and destination attractiveness. Jhikargachha experienced a road density increase of about 689% and a night-light intensity increase of nearly 195%. Similarly, Sharsha recorded a road density increase of approximately 634% along with a 160% increase in night-light intensity. As a result, potential accessibility increased in those areas, as seen in Figure 2.

3.1 Potential Accessibility (2015)

The analysis of potential accessibility for 2015 reveals a clear spatial inequality among the 58 upazilas within the study area. Higher Accessibility were was strongly concentrated around major urban and economically active regions, while remote coastal and environmentally constrained areas experienced very low accessibility. The results indicate that pre-Padma Bridge accessibility in southwest Bangladesh was heavily influenced by road connectivity, economic concentration, and travel impedance caused by river barriers and limited transportation infrastructure.

Rank	Upazila	District	Potential Accessibility
1	Daulatpur	Khulna	1.309
2	Jessore	Jessore	1.143

3	Rupsa	Khulna	1.101
4	Faridpur	Faridpur	0.920
5	Dighalia	Khulna	0.831

Table-1: Top Five Accessible Upazila in 2015

Figure 2 demonstrates the upazilas that had relatively high accessibility in 2015 because they had better road networks, larger populations, and stronger economic activities shown by higher night-time light intensity. Urban areas around Khulna and Jessore were already important regional economic centers before the Padma Bridge opened.

Upazila like Nagarkandi, Zanjira, Chaugachha has some level of transportation connectivity and population, but has a lack of strong economic intensity and network advantages. But due to their relative geographic proximity to Dhaka, their potential accessibility scores moderate.

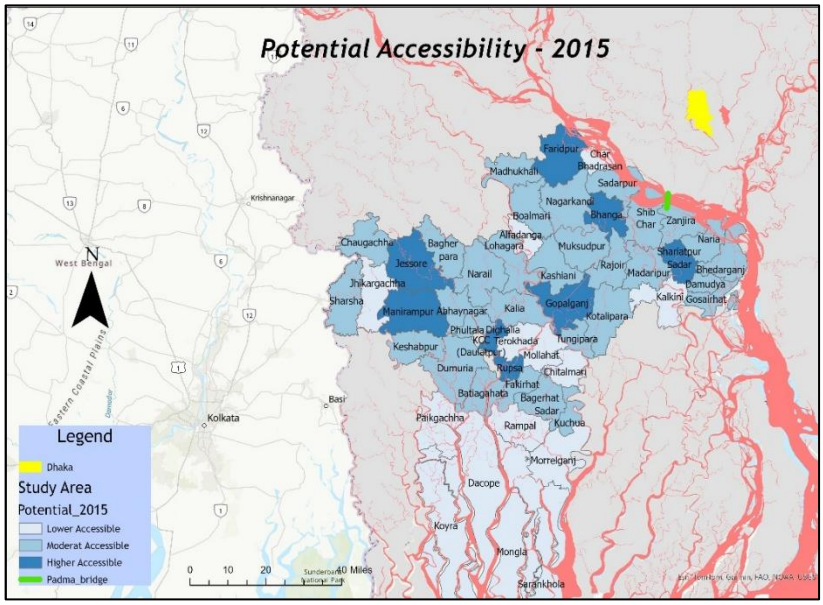


Figure-2: Potential Accessibility-2015

3.2 Potential Accessibility (2025)

Figure 3 shows the 2025 potential accessibility analysis reveals a major transformation in the spatial structure of southwest Bangladesh following the opening of the Padma Bridge corridor. Accessibility levels increased substantially across nearly all 58 upazilas. The results suggest that the reduction in travel time, combined with improvements in road connectivity and economic activity, significantly reshaped the regional accessibility hierarchy.

Rank	**Upazila**	**District**	**Potential Accessibility**
1	Daulatpur	Khulna	5.842
2	Rupsa	Khulna	5.421
3	Jessore	Jessore	5.108
4	Dighalia	Khulna	4.937
5	Faridpur	Faridpur	4.602

Table-2: Top Five Accessible Upazila in 2025

In 2025, Kotalipara, Zanjira, Chaugachha Upazilas achieved moderate accessibility improvements and benefited from improved regional connectivity after the Padma Bridge. This can be seen in Figure 3. Many previously isolated upazilas moved into the middle accessibility category because of better road access and increasing economic activity.

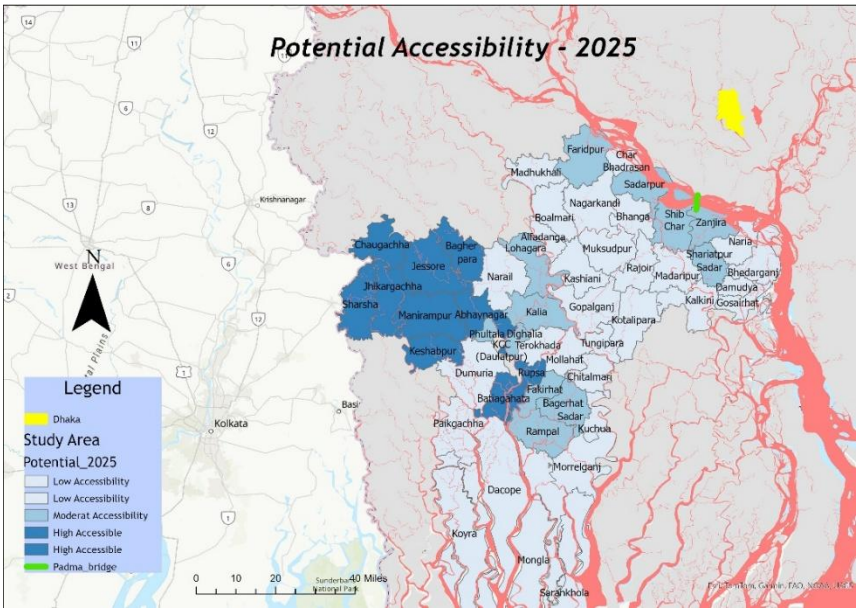


Figure-3: Potential Accessibility-2025

3.3 Change in Potential Accessibility (2015–2025)

The percentage change analysis of potential accessibility reveals the magnitude of spatial transformation across the 58 upazilas following the development of the Padma Bridge corridor. It is clear from the result that accessibility improved in every upazila, although the magnitude of improvement varied significantly across the study area. These differences highlight how the Padma Bridge is benefiting several southwestern and previously less-connected regions.

Rank	Upazila	District	Percentage Change (%)
1	Sharsha	Jessore	1248%
2	Jhikargachha	Jessore	1187%
3	Dacope	Khulna	1042%
4	Sarankhola	Bagerhat	1035%
5	Mongla	Bagerhat	986%

Table-3: Ranked Changes in Potential Accessibility (2015–2025)

The upazilas in Figure 4 experienced extremely large accessibility gains because of the combined effects of reduced travel time, rapid road network expansion, and increased economic activity represented by night-time light intensity. Several of these areas previously had relatively low accessibility in 2015.

Daulatpur, Rupsa, Jessore these upazilas recorded the lowest percentage increases, although they still experienced substantial accessibility growth in absolute terms. These areas already had relatively high accessibility levels in 2015, meaning that even large absolute improvements resulted in smaller proportional percentage changes.

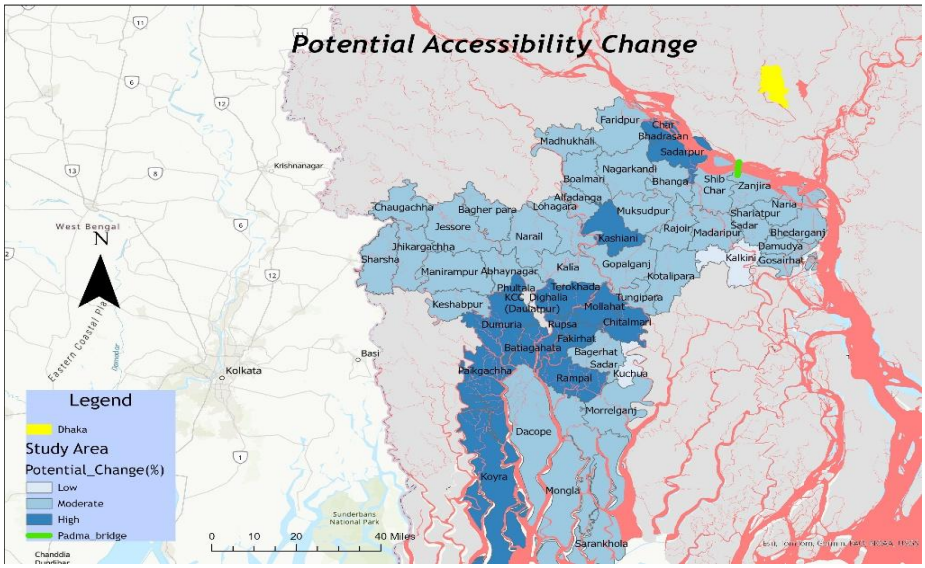


Figure-4: Potential accessibility change

4. Discussion

The quantitative findings in this project using the Hansen gravity model provide robust, empirical support for the initial thesis: the Padma Corridor works as a spatial transformer for southern Bangladesh. The massive percentage increases in potential accessibility—particularly in the Jessore and Khulna regions—validate that the bridge is not just a physical transport link, but an economic multiplier.

Moreover, this newly mapped accessibility strongly supports the possibility of urban decentralization. Major industries, such as the Ready-Made Garment (RMG) sector has no other option but to setup in and around the dense capital of Dhaka to maintain reliable access to national and international supply chains and labor force. As the travel times to the southwest drastically reduced, regional industrial center in Khulna and the emerging economic hubs in Jessore now offer geographically and economically viable alternatives.

It is for clear that transformation infrastructure increase accessibility, but that does not necessarily translate into economic development. It may cause: redistribution of economic activity rather than net growth. Transport investment alone does not guarantee economic development, it may work as a supporting factor rest will depends on economic, institutional, and political conditions and also the timing of investment (Banister & Berechman, 2001).

5. Conclusion

In conclusion, this study used the Hansen gravity model to measure the spatial transformation of southern Bangladesh caused by the Padma Bridge. By combining remote sensing data, such as VIIRS night-time light intensity, with road connectivity and travel-time data, the study identified a major increase in potential accessibility between 2015 and 2025.

However, this study proves that the Padma Corridor has fundamentally altered the economic geography of the region. Travel from the capital to southern maritime hubs like Mongla Port is now reliable and fast. This makes the southwest part a highly viable alternative for land-intensive manufacturing and commercial investment. This crucial transportation infrastructure should utilize to achieve a more sustainable and balanced development.

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Appendix:

I used ChatGPT as a support tool during development of this annotated bibliography. I used AI primarily for brainstorming ideas, clarification of complex concepts related to topic and as a proof reader to check grammar and spelling. All final writing were completed in my own words. Example Prompts Used

“What are key foundational papers on accessibility in transportation planning?”

“Suggest few paper where theory of accessibility is applied.”

“Explain Hansen (1959) accessibility model in simple terms.”

“Proof read blew paragraph for grammar and spelling?”

However, I did not directly copy AI-generated text into the final submission. Instead, I used the information as guidance and rewrote all content in my own words. ChatGPT was helpful in quickly summarizing complex academic ideas and suggesting relevant literature.